EXPERIMENT 20: TEST FOR ANALGESIA (MOUSE WRITHING ASSAY)

300 Level Pharmacology Practical & In-Course Examination Master Study Guide

Department of Pharmacology, College of Medicine / University Teaching Hospital
Source Document: Practical Lab Manual & Report — Page 112 (Printed Experimental Text)
Target: 30 Verbatim High-Yield Multiple Choice Questions (MCQs) & In-Course Rationales

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📄 Exact Verbatim Transcription of Page 112 (Printed Text)

For complete reference and verbatim verification, here is the exact text printed on Page 112 of the laboratory manual:

EXPERIMENT 20

TEST FOR ANALGESIA: Mouse Writhing Assay

Drugs: Acetic acid (0.6%), morphine sulphate, acetylsalicylic acid, unknown drugs (plant extracts).
Animals: Albino mice
Others: Syringes & needles, stop watches, weighing balance, oral cannula
Procedure: Randomly divide the animals into groups and treat as follows:

Group I: Distilled water - 10ml/kg; p.o
Group II: Acetylsalicylic acid - 100mg/kg; p.o
Group III: Morphine 10mg/kg; s.c
Group IV: Unknown 50mg/kg; p.o
Group V: Unknown 100mg/kg; p.o
Group VI: Unknown 200mg/kg; p.o

After 30mins, inject acetic acid (10ml/kg) i.p, to each mouse and note the number of writhes (a syndrome characterized by a wave of contraction of the abdominal musculature, followed by extension of hind lings) every 5mins. interval for 30 mins. A reduction in the writhing number, as compared with the control, is considered evident of analgesia.

$%\text{ Inhibition} = \frac{\text{Mean (writhes, control)} - \text{Mean (writhes, drug)}}{\text{Mean (writhes, control)}} × 100

QUESTIONS
1. Suggest a possible mechanism of action (peripheral or central) for the unknown drug.
2. What can you deduce about the onset and duration of action of each of the drugs and unknown, concerning the analgesic effect.
3. Plot a graph of % inhibition vs log dose.

🎯 High-Yield Conceptual Summary for In-Course Exams

1. **Pathophysiology of the Mouse Writhing Assay (Koster's Test / Witkin's Method):**
   - **Noxious Stimulus:** Intraperitoneal injection of **0.6% acetic acid** induces acute localized chemical peritonitis and visceral peritoneal inflammation.
   - **Pain Mediators Released:** Acetic acid triggers endogenous release of **prostaglandins (ext{PGE}*2, ext{PGF}*{2alpha})**, bradykinin, histamine, serotonin, substance P, and pro-inflammatory cytokines (ext{TNF}-alpha, ext{IL}-1□eta, ext{IL}-8) into the peritoneal fluid.
   - **Nociceptor Sensitization:** Prostaglandins sensitize primary visceral nociceptive afferent C-fibers, lowering their firing threshold.
2. **Behavioral Definition of a "Writhe":**
   - Characterized by a **wave of contraction of the abdominal musculature, elongation of the body, and backward extension of the hind limbs** (printed as "*extension of hind lings*").
3. **Reference Standards (Central vs Peripheral):**
   - **Acetylsalicylic Acid (Aspirin - 100 mg/kg p.o.):** Peripherally acting reference standard. Irreversibly acetylates and inhibits cyclooxygenase enzymes (COX-1 and COX-2) ightarrow stops local prostaglandin synthesis ightarrow prevents peritoneal nociceptor sensitization.
   - **Morphine Sulphate (10 mg/kg s.c.):** Centrally acting reference standard. Agonist at mu-opioid receptors in the periaqueductal gray, spinal cord dorsal horn, and rostral ventromedial medulla ightarrow inhibits substance P release and hyperpolarizes ascending pain projection neurons.
4. **Critical Exam Pitfall (Question 1 Mechanism):**
   - The mouse writhing test is a **non-specific algesic model**: it responds to **BOTH centrally acting analgesics (opioids)** and **peripherally acting analgesics (NSAIDs)**!
   - *How to differentiate in an exam:* To confirm whether a plant extract acts centrally or peripherally, it must be evaluated alongside a **thermal nociceptive model** (such as the Hot Plate or Tail Flick test). Centrally acting opioids inhibit both thermal and chemical pain; peripherally acting NSAIDs inhibit chemical writhing but are **ineffective against acute thermal pain**!
5. **Pre-treatment Interval (30 Minutes):**
   - Acetic acid is injected **30 minutes after drug administration** to allow complete gastrointestinal absorption and systemic distribution of oral agents (distilled water, aspirin, unknown extract) and subcutaneous morphine.

📝 The 30 Verbatim In-Course Multiple Choice Questions (MCQs)

Question 1: Experiment Number & Designation

What is the official numerical designation and title of this laboratory experiment as printed at the top of Page 112?

- A) Experiment 3: Individual Variations in the Response to Drugs

- B) Experiment 5: Effect of Route of Administration on Drug Effect
- C) Experiment 20: Test for Analgesia: Mouse Writhing Assay
- D) Experiment 22: Evaluation of Local Anesthetic Potency by Infiltration

Correct Answer: `C`

- **Verbatim Text on Page 112:** `EXPERIMENT 20 / TEST FOR ANALGESIA: Mouse Writhing Assay`
- **In-Course Explanation:** In practical spot tests and vivas, examiners ask students to state the exact experiment title and number. Experiment 20 is the classical mouse writhing model for screening analgesic properties.
- **Distractor Breakdown:** Option A is Experiment 3; Option B is Experiment 5; Option D describes surface/infiltration local anesthesia bioassays.

Question 2: Chemical Algesic Agent Used

Which chemical agent is listed under the "Drugs" heading on Page 112 to induce visceral pain and writhing behavior?

- A) Formalin
- B) Acetic acid (0.6%)
- C) Capsaicin
- D) Carrageenan

Correct Answer: `B`

- **Verbatim Text on Page 112:** `Drugs: Acetic acid (0.6%), morphine sulphate, acetylsalicylic acid, unknown drugs (plant extracts).`
- **In-Course Explanation:** Dilute acetic acid (0.6%) is the standard chemical irritant injected intraperitoneally to trigger acute chemical peritonitis and abdominal writhing.
- **Distractor Breakdown:** Option A (formalin) is used in the two-phase formalin paw licking test; Option C (capsaicin) stimulates TRPV1 channels; Option D (carrageenan) induces acute paw edema in anti-inflammatory tests.

Question 3: Exact Acetic Acid Concentration

What is the exact percentage concentration of the acetic acid solution specified on Page 112?

- A) 0.1%
- B) 0.6%
- C) 2.0%
- D) 5.0%

Correct Answer: `B`

- **Verbatim Text on Page 112:** `Acetic acid (0.6%)`
- **In-Course Explanation:** 0.6% v/v acetic acid in distilled water provides a reproducible, non-lethal, self-limiting visceral painful stimulus that lasts approximately 30 minutes.

- **Distractor Breakdown:** Options A, C, and D are incorrect concentrations; higher concentrations (e.g., 5%) cause chemical tissue necrosis and shock.

Question 4: Centrally Acting Reference Standard

Which drug listed under "Drugs" serves as the centrally acting opioid analgesic reference standard?

- A) Paracetamol
- B) Morphine sulphate
- C) Naloxone hydrochloride
- D) Pethidine

Correct Answer: `B`

- **Verbatim Text on Page 112:** `morphine sulphate`
- **In-Course Explanation:** Morphine sulphate is the prototype opioid analgesic acting centrally via mu-opioid receptors to suppress nociceptive transmission.
- **Distractor Breakdown:** Option A is a weak non-opioid antipyretic; Option C is an opioid receptor antagonist; Option D is a synthetic opioid not listed on Page 112.

Question 5: Peripherally Acting Reference Standard

Which agent listed on Page 112 serves as the peripherally acting non-steroidal anti-inflammatory reference standard?

- A) Acetylsalicylic acid
- B) Ibuprofen
- C) Indomethacin
- D) Celecoxib

Correct Answer: `A`

- **Verbatim Text on Page 112:** `acetylsalicylic acid`
- **In-Course Explanation:** Acetylsalicylic acid (aspirin) is the classical peripheral analgesic standard that inhibits cyclooxygenase (COX-1/COX-2), blocking local prostaglandin synthesis in the peritoneum.
- **Distractor Breakdown:** Options B, C, and D are other NSAIDs, but acetylsalicylic acid is the exact agent specified in the manual.

Question 6: Nature of the Unknown Test Drug

How does Page 112 describe the nature and biological source of the "unknown drugs"?

- A) Synthetic benzodiazepines
- B) plant extracts
- C) Semisynthetic penicillins
- D) Heavy metal chelators

Correct Answer: `B`

- **Verbatim Text on Page 112:** `unknown drugs (plant extracts).`
- **In-Course Explanation:** In pharmacology academic practicals, students evaluate crude botanical/phytochemical extracts (unknown plant extracts) for prospective analgesic efficacy against modern clinical standards.
- **Distractor Breakdown:** Options A, C, and D represent unrelated chemical and therapeutic classes.

Question 7: Animal Model Specified

What specific laboratory animal species and strain is designated on Page 112?

- A) Wistar rats
- B) Albino mice
- C) Dunkin-Hartley guinea pigs
- D) Golden Syrian hamsters

Correct Answer: `B`

- **Verbatim Text on Page 112:** `Animals: Albino mice`
- **In-Course Explanation:** Albino mice (*Mus musculus*) are universally used for writhing assays because of their compliant abdominal wall, low cost, and distinct behavioral writhing response.
- **Distractor Breakdown:** Option A (rats) was used in Experiment 5; Options C and D are not used for writhing assays.

Question 8: Specialized Oral Dosing Apparatus

Which instrument listed under "Others" is used specifically to administer oral drugs to the mice?

- A) Oral cannula
- B) Tuberculin pipette
- C) Glass dropper
- D) Automated syringe pump

Correct Answer: `A`

- **Verbatim Text on Page 112:** `Others: Syringes & needles, stop watches, weighing balance, oral cannula`
- **In-Course Explanation:** An oral cannula (curved ball-tipped gavage needle) is gently advanced through the oropharynx into the stomach, ensuring reliable delivery without injuring the esophageal mucosa.
- **Distractor Breakdown:** Options B and C allow regurgitation and tracheal aspiration; Option D is for continuous parenteral infusions.

Question 9: Time Measurement Instrument

What timing device is explicitly listed in the plural under "Others" on Page 112?

- A) Hourglasses
- B) stop watches
- C) Metronomes
- D) Kymograph clocks

Correct Answer: `B`

- **Verbatim Text on Page 112:** `stop watches`
- **In-Course Explanation:** Multiple stopwatches are required simultaneously to track the 30-minute pre-treatment period and count writhes across consecutive 5-minute intervals.
- **Distractor Breakdown:** Options A, C, and D lack the required fractional-second precision.

Question 10: Gravimetric Instrument

Which instrument listed under "Others" is used to weigh the mice prior to dose calculation?

- A) Spring balance
- B) weighing balance
- C) Analytical beam balance
- D) Hydrometer

Correct Answer: `B`

- **Verbatim Text on Page 112:** `weighing balance`
- **In-Course Explanation:** A sensitive electronic or digital weighing balance measures individual mouse weights (typically 18–30 g) to allow exact volume calculations in ext{mL}.
- **Distractor Breakdown:** Option A is inaccurate for small rodents; Option D measures fluid specific gravity.

Question 11: Experimental Randomization Requirement

According to the Procedure on Page 112, how must the test animals be allocated into treatment groups?

- A) Sequentially by decreasing body weight
- B) Randomly divide the animals into groups
- C) Based on baseline motor activity
- D) By selecting all males into one group and females into another

Correct Answer: `B`

- **Verbatim Text on Page 112:** `Procedure: Randomly divide the animals into groups and treat as follows:`
- **In-Course Explanation:** Random allocation prevents selection bias and ensures confounding factors (age, basal nociceptive threshold) are distributed equally across groups.
- **Distractor Breakdown:** Options A, C, and D introduce systematic bias.

Question 12: Total Number of Treatment Groups

How many distinct experimental treatment groups are formally listed in the Procedure on Page 112?

- A) 3 groups
- B) 4 groups
- C) 6 groups
- D) 10 groups

Correct Answer: `C`

- **Verbatim Text on Page 112:** `Group I ... Group II ... Group III ... Group IV ... Group V ... Group VI`
- **In-Course Explanation:** There are 6 treatment groups: Group I (negative control), Group II (aspirin standard), Group III (morphine standard), and Groups IV–VI (three graded doses of unknown plant extract).
- **Distractor Breakdown:** Options A, B, and D do not match the six roman-numeral groups printed on Page 112.

Question 13: Group I Treatment & Dosage

What substance, dose, and route of administration are specified for Group I on Page 112?

- A) Normal saline 5ml/kg; i.p
- B) Distilled water - 10ml/kg; p.o
- C) 5% Dextrose 10ml/kg; p.o
- D) Liquid paraffin 2ml/kg; p.o

Correct Answer: `B`

- **Verbatim Text on Page 112:** `Group I: Distilled water - 10ml/kg; p.o`
- **In-Course Explanation:** Distilled water given orally at 10 mL/kg serves as the negative control (vehicle control).
- **Distractor Breakdown:** Options A, C, and D are not listed for Group I on Page 112.

Question 14: Pharmacological Function of Group I

In the context of bioassay design, what is the indispensable scientific role of Group I?

- A) Positive standard for opioid analgesia
- B) Negative (vehicle) control to establish the maximum baseline writhing frequency without drug inhibition
- C) Solvent control to demonstrate vehicle toxicity
- D) Baseline control for thermal pain threshold

Correct Answer: `B`

- **In-Course Explanation:** Group I provides the reference baseline writhing count (ext{Mean}_{ ext{control}}) required in the denominator and numerator of the percentage inhibition equation.
- **Distractor Breakdown:** Option A describes morphine; Option C is for potentially toxic non-aqueous solvents; Option D applies to hot plate tests.

Question 15: Group II Treatment & Dosage

What drug, dose, and route are specified for Group II on Page 112?

- A) Acetylsalicylic acid - 100mg/kg; p.o
- B) Morphine 10mg/kg; s.c
- C) Ibuprofen 50mg/kg; p.o
- D) Paracetamol 100mg/kg; i.p

Correct Answer: `A`

- **Verbatim Text on Page 112:** `Group II: Acetylsalicylic acid - 100mg/kg; p.o`
- **In-Course Explanation:** Group II receives 100 mg/kg acetylsalicylic acid orally, establishing the benchmark for peripheral COX inhibition.
- **Distractor Breakdown:** Option B is Group III; Options C and D are alternative drugs not listed in this protocol.

Question 16: Group III Treatment & Dosage

What drug, dose, and route are specified for Group III on Page 112?

- A) Acetylsalicylic acid - 100mg/kg; p.o
- B) Morphine 10mg/kg; s.c
- C) Codeine 30mg/kg; p.o
- D) Fentanyl 0.1mg/kg; i.v

Correct Answer: `B`

- **Verbatim Text on Page 112:** `Group III: Morphine 10mg/kg; s.c`
- **In-Course Explanation:** Group III receives 10 mg/kg morphine sulphate subcutaneously.
- **Distractor Breakdown:** Option A is Group II; Options C and D are alternative opioids not listed on Page 112.

Question 17: Pharmacological Rationale for Morphine Route (s.c.)

Why is Morphine administered subcutaneously (s.c.) in Group III while Acetylsalicylic Acid and the Unknowns are given orally (p.o.)?

- A) Morphine is insoluble in water
- B) Morphine undergoes extensive first-pass hepatic metabolism and has poor, erratic oral bioavailability
- C) Subcutaneous injection permanently prevents morphine tolerance

- D) Morphine is destroyed by gastric amylase

Correct Answer: `B`

- **In-Course Explanation:** Morphine has low and variable oral bioavailability (~20–40%) in rodents and humans due to prominent presystemic hepatic glucuronidation (to morphine-3-glucuronide). Subcutaneous injection bypasses the portal circulation, ensuring rapid, predictable systemic absorption.
- **Distractor Breakdown:** Option A is false (morphine sulphate is water-soluble); Option C is false (tolerance still develops); Option D is incorrect (amylase digests starch, not alkaloids).

Question 18: Group IV Treatment (Lowest Unknown Dose)

What treatment, dose, and route are specified for Group IV on Page 112?

- A) Unknown 10mg/kg; p.o
- B) Unknown 50mg/kg; p.o
- C) Unknown 100mg/kg; p.o
- D) Unknown 50mg/kg; i.p

Correct Answer: `B`

- **Verbatim Text on Page 112:** `Group IV: Unknown 50mg/kg; p.o`
- **In-Course Explanation:** Group IV evaluates the lowest dose of the unknown plant extract (50 mg/kg) via the oral route.
- **Distractor Breakdown:** Option A is a subtherapeutic dose; Option C is Group V; Option D specifies an incorrect route (i.p. instead of p.o.).

Question 19: Group V Treatment (Intermediate Unknown Dose)

What treatment, dose, and route are specified for Group V on Page 112?

- A) Unknown 50mg/kg; p.o
- B) Unknown 100mg/kg; p.o
- C) Unknown 200mg/kg; p.o
- D) Unknown 100mg/kg; s.c

Correct Answer: `B`

- **Verbatim Text on Page 112:** `Group V: Unknown 100mg/kg; p.o`
- **In-Course Explanation:** Group V evaluates the intermediate dose (100 mg/kg p.o.), matching the dose used for the aspirin standard (100 mg/kg p.o.).
- **Distractor Breakdown:** Option A is Group IV; Option C is Group VI; Option D specifies an incorrect route.

Question 20: Group VI Treatment (Highest Unknown Dose)

What treatment, dose, and route are specified for Group VI on Page 112?

- A) Unknown 100mg/kg; p.o

- B) Unknown 200mg/kg; p.o
- C) Unknown 400mg/kg; p.o
- D) Unknown 500mg/kg; p.o

Correct Answer: `B`

- **Verbatim Text on Page 112:** `Group VI: Unknown 200mg/kg; p.o`
- **In-Course Explanation:** Group VI evaluates the highest dose (200 mg/kg p.o.) to establish a dose-dependent graded increase in % inhibition.
- **Distractor Breakdown:** Option A is Group V; Options C and D are not listed in the printed protocol.

Question 21: Mathematical Progression of Unknown Doses

What mathematical relationship characterizes the dose escalation of the unknown drug across Groups IV, V, and VI?

- A) Linear progression (50, 60, 70 mg/kg)
- B) Arithmetic doubling / geometric 2-fold progression (50, 100, 200 mg/kg)
- C) Logarithmic 10-fold progression (1, 10, 100 mg/kg)
- D) Exponential Fibonacci series (50, 100, 150 mg/kg)

Correct Answer: `B`

- **In-Course Explanation:** The doses (50 ightarrow 100 ightarrow 200 mg/kg) double at each step, yielding equally spaced increments when plotted on a logarithmic axis (log_{10} 50 = 1.70, log_{10} 100 = 2.00, log_{10} 200 = 2.30).
- **Distractor Breakdown:** Options A, C, and D do not match the 50, 100, 200 mg/kg sequence.

Question 22: Pre-treatment Waiting Period

How much time must elapse after drug administration before the acetic acid challenge is injected?

- A) Immediately (0 minutes)
- B) After 10mins
- C) After 30mins
- D) After 2 hours

Correct Answer: `C`

- **Verbatim Text on Page 112:** `After 30mins, inject acetic acid (10ml/kg) i.p, to each mouse`
- **In-Course Explanation:** A 30-minute absorption window ensures oral drugs cross the gastrointestinal mucosa and achieve active concentrations in plasma and peritoneal tissue before the noxious stimulus begins.
- **Distractor Breakdown:** Options A and B are too short for oral absorption; Option D risks significant metabolic elimination.

Question 23: Acetic Acid Dosage & Route

What exact dose volume and route of administration are specified for the acetic acid challenge on Page 112?

- A) 1ml/kg; s.c
- B) 10ml/kg i.p
- C) 5ml/kg; p.o
- D) 0.5ml/kg; i.v

Correct Answer: `B`

- **Verbatim Text on Page 112:** `inject acetic acid (10ml/kg) i.p, to each mouse`
- **In-Course Explanation:** 10 mL/kg of 0.6% acetic acid is injected into the peritoneal cavity (0.1 mL per 10 g mouse weight).
- **Distractor Breakdown:** Options A, C, and D specify incorrect volumes and non-peritoneal routes.

Question 24: Verbatim Definition of a "Writhe"

How is a "writhe" specifically defined on Page 112?

- A) Rapid scratching of the snout with the forepaws followed by head shakes
- B) a syndrome characterized by a wave of contraction of the abdominal musculature, followed by extension of hind lings
- C) Sudden clonic convulsions with loss of righting reflex
- D) Continuous circular ambulation around the perimeter of the cage

Correct Answer: `B`

- **Verbatim Text on Page 112:** `( a syndrome characterized by a wave of contraction of the abdominal musculature, followed by extension of hind lings)`
- **In-Course Explanation:** *High-yield verbatim detail:* The manual explicitly defines writhing as abdominal muscle contraction followed by backward hind-limb extension (note the exact printed typo: "hind lings").
- **Distractor Breakdown:** Option A describes facial formalin pain; Option C describes pentylenetetrazol seizures; Option D describes rotational stereotypy.

Question 25: Writhing Observation Protocol

According to Page 112, over what total duration and at what specific intervals must writhes be counted?

- A) Continuously for 5 minutes only
- B) every 5mins. interval for 30 mins.
- C) Every 15 minutes for 2 hours
- D) Once at the 60-minute mark for 10 minutes

Correct Answer: `B`

- **Verbatim Text on Page 112:** `every 5mins. interval for 30 mins.`
- **In-Course Explanation:** Observations are divided into six 5-minute blocks (0–5, 5–10, 10–15, 15–20, 20–25, 25–30 min) over 30 minutes, capturing both onset and peak writhing intensity.
- **Distractor Breakdown:** Options A, C, and D do not follow the printed schedule.

Question 26: Stated Criterion for Analgesia

What exact criterion on Page 112 is considered proof of analgesic efficacy?

- A) Complete cessation of spontaneous locomotion
- B) A reduction in the writhing number, as compared with the control, is considered evident of analgesia.
- C) An increase in pupil diameter of more than 50%
- D) Elevation of body temperature by 1.5°C

Correct Answer: `B`

- **Verbatim Text on Page 112:** `A reduction in the writhing number, as compared with the control, is considered evident of analgesia.`
- **In-Course Explanation:** Analgesia is quantified by a statistically significant reduction in total writhes in treated mice relative to vehicle controls.
- **Distractor Breakdown:** Option A indicates sedation or motor deficit; Option C is mydriasis; Option D is pyrexia.

Question 27: Mathematical Equation for Percentage Inhibition

Which mathematical formula is printed on Page 112 to calculate the percentage inhibition of writhing?

- A) % ext{ Inhibition} = rac{ ext{Mean}*{ ext{drug}}}*{ ext{Mean}*{ ext{control}}} imes 100
- B) % ext{ Inhibition} = rac{ ext{Mean (writhes, control)} - ext{Mean (writhes, drug)}}{ ext{Mean (writhes, control)}} imes 100
- C) % ext{ Inhibition} = rac{ ext{Mean}*{ ext{control}} - ext{Mean}*{ ext{drug}}}{ ext{Mean}_{ ext{drug}}} imes 100
- D) % ext{ Inhibition} = (ext{Mean}*{ ext{control}} + ext{Mean}*{ ext{drug}}) imes 100

Correct Answer: `B`

- **Verbatim Text on Page 112:**
  % ext{ Inhibition} = rac{ ext{Mean (writhes, control)} - ext{Mean (writhes, drug)}}{ ext{Mean (writhes, control)}} imes 100
- **In-Course Explanation:** *Essential calculation formula:* Computes the fractional reduction in writhing relative to the control group, multiplied by 100.
- **Distractor Breakdown:** Option A calculates % survival/activity; Option C places drug in the denominator (mathematically incorrect); Option D sums the means.

Question 28: First Post-Lab Question Printed on Page 112

What is the exact text of Question 1 under "QUESTIONS" on Page 112?

- A) What is the chemical structure of acetylsalicylic acid?
- B) Suggest a possible mechanism of action (peripheral or central) for the unknown drug.
- C) What is the lethal dose 50 of morphine in albino mice?
- D) How does acetic acid cause necrosis of the peritoneal mesentery?

Correct Answer: `B`

- **Verbatim Text on Page 112:** `1. Suggest a possible mechanism of action (peripheral or central) for the unknown drug.`
- **In-Course Explanation:** Focuses on deducing whether the extract acts centrally (like morphine) or peripherally (like aspirin).
- **Distractor Breakdown:** Options A, C, and D are not printed on Page 112.

Question 29: In-Course Viva Pearl on Question 1 Mechanism

In answering Question 1, why can the Mouse Writhing Assay NOT definitively establish whether a drug acts exclusively centrally or peripherally on its own?

- A) Mice lack central opioid receptors in their spinal cord
- B) The assay is non-specific and is inhibited by BOTH centrally acting opioids and peripherally acting NSAIDs
- C) Acetic acid is only absorbed into skeletal muscle, bypassing the nervous system
- D) Aspirin crosses the blood-brain barrier and stimulates mu-receptors

Correct Answer: `B`

- **In-Course Explanation:** *Top examination viva trap:* The writhing assay is sensitive to both central analgesics (which suppress pain transmission in the brain/spinal cord) and peripheral analgesics (which stop local prostaglandin synthesis in the peritoneum). To prove an agent acts centrally, it must be paired with thermal models (hot plate / tail flick), which respond only to central opioids.
- **Distractor Breakdown:** Option A is false (mice have abundant opioid receptors); Option C is false (acetic acid acts locally on visceral peritoneum); Option D is false (aspirin does not activate mu-receptors).

Question 30: Third Post-Lab Question Printed on Page 112

What specific graphical analysis is required by Question 3 on Page 112?

- A) Plot a Lineweaver-Burk double reciprocal plot
- B) Plot a graph of % inhibition vs log dose.
- C) Plot a Scatchard plot of bound versus free ligand
- D) Plot plasma concentration versus clearance time

Correct Answer: `B`

- **Verbatim Text on Page 112:** `3. Plot a graph of % inhibition vs log dose.`
- **In-Course Explanation:** Plotting % Inhibition (Y-axis) versus Log Dose (X-axis) yields a sigmoidal log dose-response curve from which the median effective inhibitory dose (ext{IC}*{50} or ext{ED}*{50}) can be calculated.
- **Distractor Breakdown:** Option A is for enzyme kinetics; Option C evaluates receptor binding affinity (B_{ ext{max}}, K_d); Option D evaluates clearance kinetics.

🏆 In-Course Quick-Review Master Cheat Sheet

Parameter	Verbatim Fact on Page 112	In-Course Clinical Pearl
Experiment Number	Experiment 20	Test for Analgesia: Mouse Writhing Assay
Full Title	Test for Analgesia: Mouse Writhing Assay	Visceral chemical algesic model
Chemical Irritant	Acetic acid (0.6%)	Triggers peritonitis & local prostaglandin release
Animals	Albino mice	Standard small-rodent writhing model
Apparatus	Syringes & needles, stop watches, balance, oral cannula	Oral cannula for gavage; stopwatches for 5-min intervals
Opioid Standard	Morphine sulphate (10 mg/kg; s.c)	Centrally acting mu-opioid agonist
NSAID Standard	Acetylsalicylic acid (100 mg/kg; p.o)	Peripherally acting non-selective COX inhibitor
Unknown Drug	Unknown drugs (plant extracts)	Graded doses: 50, 100, 200 mg/kg p.o.
Vehicle Control	Group I: Distilled water (10 ml/kg; p.o)	Negative control establishing maximum writhing count
Waiting Period	After 30mins	Allows complete GI absorption of oral treatments
Challenge Dose	10 ml/kg i.p	Intraperitoneal injection of 0.6% acetic acid
Writhe Definition	Abdominal contraction + extension of hind lings	Contraction wave followed by hind-limb stretch
Counting Schedule	Every 5mins. interval for 30 mins.	Six 5-minute recording bins over 30 minutes
Analgesia Criterion	Reduction in writhing number vs control	Statistically significant decrease in total writhes
Formula	(ext{Control} - ext{Drug}) / ext{Control} imes 100	Calculates percentage inhibition of writhing
Printed Q1	Suggest peripheral or central mechanism	Assay is non-specific; needs thermal test to distinguish
Printed Q3	Plot % inhibition vs log dose	Yields sigmoidal curve to determine ext{IC}_{50}$